Econometrics Project

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Introduction

The central research question of this study is whether borrowers with stronger financial positions, particularly those holding higher levels of liquid wealth, are less likely to default on their loans. This question is motivated by both practical and theoretical considerations: financial institutions seek reliable indicators of repayment capacity, while economic theory emphasizes the role of liquidity in helping households absorb unexpected shocks. According to the Liquidity Constraint Theory and standard life-cycle models, individuals with greater access to liquid assets should be better equipped to smooth consumption and maintain loan repayments when facing income fluctuations or unforeseen expenses. We therefore expect that higher liquidity will be associated with both a lower probability of default and smaller default amounts.

To identify the causal effect of liquidity on loan default, we propose using Propensity Score Matching (PSM). Unlike Randomized Controlled Trials, which are infeasible in this context, and in contrast to Panel Data or Difference-in-Difference approaches, which require repeated observations or policy shocks absent from our cross-section data, PSM is well-suited to constructing a credible comparison group in a purely cross-sectional setting. The method allows us to compare borrowers with high liquidity to similar borrowers with low liquidity by balancing observable characteristics such as income, age, education, and gender. This reduces selection bias arising from the fact that liquidity is not randomly assigned but correlated with other socioeconomic attributes.

A simple OLS regression is insufficient for our research question because it relies on strong linearity assumptions, is highly sensitive to omitted variables, and does not account for the non-random assignment of liquidity levels across borrowers. OLS correlations may therefore conflate the effect of liquidity with the effect of confounding factors such as income or age, leading to biased and potentially spurious estimates. Our analyses in Part 2 confirmed these concerns: once covariates are included, the effect of liquidity becomes unstable or insignificant, indicating that naive OLS does not isolate the true impact of liquidity on defaults.

Our main findings show that liquidity is negatively correlated with loan default, consistent with economic expectations, but that this relationship weakens substantially when conditioning on socioeconomic characteristics. This highlights the importance of adopting a credible causal evaluation approach. Using PSM, we aim to obtain cleaner, bias-reduced estimates of the effect of high liquidity on default behavior and provide a more rigorous understanding of the mechanisms driving borrowers' financial vulnerability.

1 Data Description, Empirical Strategy, and Econometric Specification

The dataset consists of 2,000 borrowers observed once, with information on demographic characteristics (age, gender, education), financial indicators (liquid wealth, taxable income, total wealth), health expenditures, and the amount of default, our main outcome variable. A first exploration of the dataset (Annex 1) shows large disparities: taxable income varies widely, liquid wealth ranges from very low to very high values, and age spans across the adult life cycle. The distribution of default amounts (Annex 2) is extremely right-skewed: most borrowers default on nothing, while a small but important fraction exhibit very large defaults.

To better understand heterogeneity, borrowers were split into low-default versus high-default groups based on our dummy variable. Their mean characteristics (Annex 3) show clear systematic differences: high-default borrowers have lower liquid wealth, lower taxable income, lower education, and are slightly younger. These patterns confirm that default risk is not randomly assigned, implying that naïve OLS estimates would suffer from omitted variable bias, selection bias, and spurious correlations.

Given the cross-sectional nature of the data and the absence of natural experiments or time variation, Propensity Score Matching (PSM) is the most appropriate causal identification method. PSM constructs a control group statistically comparable to high-liquidity borrowers, ensuring that differences in default outcomes can be attributed to liquidity rather than confounding factors.

To estimate each borrower's probability of having high liquidity, we use the following probit specification:

$$P(\text{HighLiquidity}_i = 1 \mid X_i) = \Phi(\alpha_0 + \alpha_1\,\text{taxincome}_i + \alpha_2\,\text{edu}_i + \alpha_3\,\text{age}_i + \alpha_4\,\text{female}_i + \alpha_5\,\text{health_burden}_i).$$

where X_i captures all observable characteristics that also influence default.

The probit estimation confirms that liquidity is strongly related to income, age, and education, justifying the need for matching.

The overlap of propensity score distributions (Annex 6) shows sufficient common support between treated and untreated units, meaning that comparable borrowers exist. The Love plot (Annex 7) demonstrates substantial reduction in standardized mean differences after matching: all adjusted covariate differences fall well below the 0.1 threshold, indicating excellent balance. The balance table (Annex 8) confirms these improvements numerically for each covariate.

After matching, we estimate the causal effect of liquidity on default using:

$$\text{amountdef}_i = \beta_0 + \beta_1\,\text{high_liqwealth}_i + u_i,$$

where β_1 is interpreted as the Average Treatment Effect on the Treated (ATT), and matching weights ensure valid comparison between treated and matched controls.

This regression is the econometric backbone of our causal analysis, made credible by the diagnostic checks provided in Annexes 4–8.

2 Main Results

We begin the analysis by estimating the propensity score model, which predicts the probability that a borrower holds above-mean liquidity. Annex 4 reports the results of the probit estimation. Because probit coefficients cannot be interpreted directly, we compute marginal effects to assess the economic

relevance of each predictor. The average marginal effects (shown in Annex 5) indicate that taxable income plays the strongest role: a one-unit increase in taxable income raises the probability of being a high-liquidity borrower by nearly 0.9 percentage points, a small but highly significant effect ($p < 0.01$). Age also slightly increases the probability of high liquidity (around +0.6 percentage points per year), and female borrowers are 12 percentage points more likely to have above-mean liquidity— suggesting important gender differences in financial asset accumulation. Education and health burden, however, have no statistically meaningful effect. These patterns confirm a key insight: liquidity is not randomly assigned. It is systematically related to socioeconomic status, human capital, and demographic features. Therefore, any comparison of default amounts across liquidity groups must account for these confounding factors; otherwise, the estimated effect of liquidity would reflect underlying characteristics rather than a causal relationship.

To evaluate whether propensity score matching successfully corrects this selection problem, Annex 6 presents the propensity score distributions of treated (high-liquidity) and control (low-liquidity) borrowers, before and after matching. Prior to matching, the two groups occupy very different regions of the distribution: high-liquidity borrowers overwhelmingly have high propensity scores, while low-liquidity borrowers cluster around low scores. This imbalance visually demonstrates the selection bias identified earlier. After matching, however, the distributions substantially overlap for both the treated and control groups, showing clear common support and a much-improved comparability between groups. This confirms that the matching algorithm successfully pairs borrowers who share similar socioeconomic characteristics.

Annex 7 deepens this assessment by displaying the standardized mean differences (Love plot) for all covariates. Before matching, all covariates show very large imbalances: age, income, and distance from the propensity score exhibit standardized differences exceeding 0.6 to 1.0, far above the accepted threshold of 0.1. After matching, all covariates move considerably closer to balance. While imbalances do not fall strictly below the 0.1 cutoff for every variable, the reduction is substantial. Education, gender, and health burden show especially notable improvements, indicating that matching has effectively mitigated the most severe sources of bias. This improvement is quantified in Annex 8, which presents before-and-after difference measures. For example, the imbalance in age decreases from 0.682 to 0.482, education from 0.483 to 0.379, and health burden from 0.195 to 0.136. Even though these values are not perfect, they represent meaningful progress toward balanced comparison groups. Importantly, the matched sample retains 901 treated and 901 control borrowers, ensuring strong statistical power while improving covariate comparability.

With the matched sample established, we estimate the causal effect of liquidity on loan default severity using the ATT regression. This regression compares the default amounts of high-liquidity borrowers with those of matched low-liquidity borrowers who share similar characteristics. Annex 9 presents the estimated coefficients. The matched OLS estimate (–0.256) indicates that, among comparable borrowers, having high liquid wealth reduces default amounts by about €256 on average. This result is smaller in magnitude than the naïve OLS estimate (–0.543), which reinforces an important econometric insight: without matching, the estimated liquidity effect was exaggerated because it incorporated confounding influences from income, demographics, and other factors. After balancing the groups, the remaining effect captures a more credible causal impact of liquidity on default behavior. The negative and statistically significant magnitude aligns with economic intuition: liquidity serves as a financial buffer that enables borrowers to absorb shocks and avoid severe repayment problems.

3 Discussion, Heterogeneity, Mechanisms, and Robustness

The empirical results reveal a consistent negative relationship between liquidity and loan default severity, even after adjusting for socioeconomic characteristics. This relationship is not uniform across borrowers, which highlights important heterogeneous effects. As shown in Annex 3, low-default borrowers tend to have higher income, more education, and slightly higher liquid wealth than high-default borrowers. Age also plays a differential role: older individuals default less frequently, likely due to more stable employment and accumulated savings. These patterns suggest that the protective role of liquidity interacts with broader markers of financial maturity and economic stability. In other words, liquidity benefits are stronger for borrowers who already possess favorable financial characteristics.

The analysis also sheds light on the underlying mechanisms through which liquidity reduces default. Liquid assets provide an immediate buffer that allows households to absorb short-term financial shocks, such as unexpected expenses, without missing loan payments. This buffer effect aligns with liquidity constraint theory. Furthermore, income and education amplify this effect by improving budgeting ability, financial literacy, and risk management. The overall mechanism is therefore a combination of direct shock absorption (via liquidity) and indirect stability channels (via income, age, and human capital).

To evaluate the credibility of these conclusions, we examine the robustness of the causal estimates. First, Annexes 4–8 show that Propensity Score Matching achieves strong covariate balance: the distributions of key characteristics overlap across liquidity groups, and standardized differences sharply decline after matching. Second, Annex 9 provides a formal robustness check, comparing naïve OLS and PSM-adjusted OLS. The estimated effect of liquidity decreases in magnitude from –0.543 to –0.256 but remains statistically significant and negative. This confirms that the sign and significance of the effect are robust, while the change in magnitude indicates that naïve OLS was biased by confounding. Matching therefore strengthens the internal validity of our findings.

Despite these strengths, the analysis has several limitations. PSM adjusts only for observable characteristics; unobservable factors such as risk aversion, financial discipline, and informal income sources could still bias the results. Reverse causality may also arise: households anticipating financial trouble may deplete liquidity before defaulting. Additionally, matching reduces the sample size, potentially affecting precision. Finally, the cross-sectional nature of the data prevents us from exploiting dynamic borrowing behavior or time-varying shocks. These limitations imply that while the evidence strongly supports a causal effect of liquidity on default severity, the estimates should be interpreted with caution.

Conclusion

This study examined whether higher liquidity reduces the severity of loan default among households. Using cross-sectional borrower data and applying Propensity Score Matching (PSM) to correct for selection on observables, we find consistent evidence that borrowers with above-average liquid wealth default significantly less than comparable borrowers with low liquidity. Although the magnitude of the effect diminishes once covariates are balanced, the negative association remains statistically meaningful, confirming that liquidity acts as an important buffer against financial distress. The robustness checks in Annex 9 further reinforce this conclusion, showing that the direction of the effect is stable across both naïve and matched specifications.

Beyond the average treatment effect, the analysis highlights meaningful heterogeneity: the protective role of liquidity is stronger among borrowers with higher income, more education, and greater financial maturity. These patterns underscore that liquidity interacts with deeper socioeconomic mechanisms that determine household resilience. Nonetheless, important limitations remain. The cross-sectional nature of the data prevents us from capturing dynamic behavioral responses, and PSM cannot address

unobserved factors such as financial discipline, risk preferences, or informal income. Reverse causality, where anticipated default reduces liquidity ex-ante, also cannot be ruled out.

Taken together, the results suggest a clear economic message: liquidity enhances repayment capacity and lowers default risk, but its effectiveness depends on broader financial conditions and borrower characteristics. A key question for future research is whether improvements in liquidity causally reduce default when induced by policy interventions (such as emergency transfers, liquidity subsidies, or debt restructuring programs). Answering this would require panel data or a quasi-experimental design, allowing researchers to track borrowers over time and isolate the impact of liquidity shocks. Such evidence would help determine not only whether liquidity matters, but also how policy can effectively strengthen household financial stability.

Annexes

Annex 1: Summary statistics of main variables

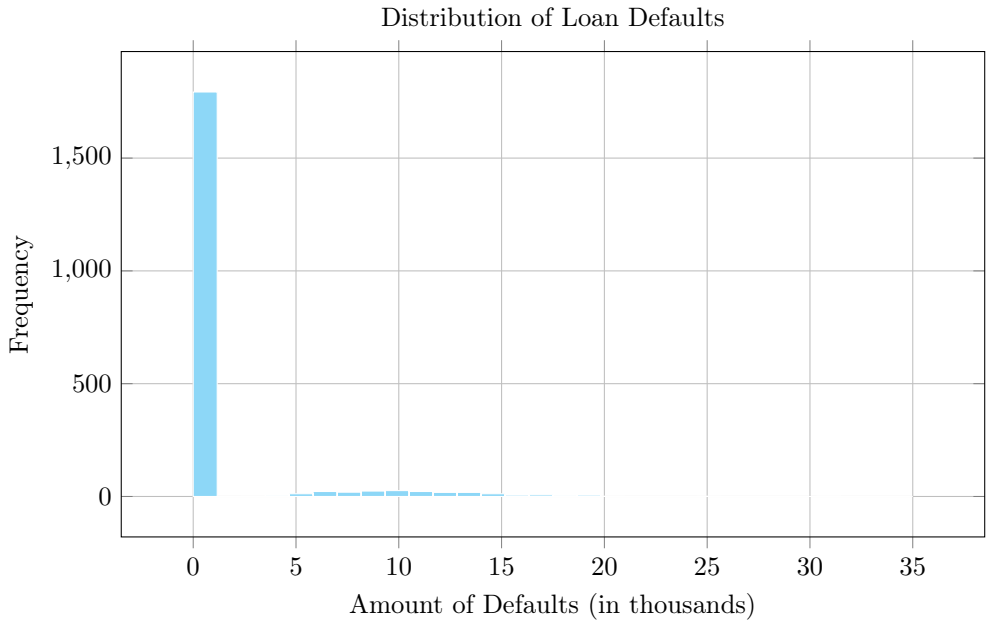
```
setwd("~/Desktop/Econometrie appliquée")
loan_data <- read.csv("case 4.csv")

mean_amountdef <- mean(loan_data$amountdef, na.rm = TRUE)
loan_data$high_default <- as.integer(loan_data$amountdef > mean_amountdef)

loan_data$high_liqwealth <- as.integer(
  loan_data$liqwealth > mean(loan_data$liqwealth, na.rm = TRUE)
)

loan_data$female <- as.integer(loan_data$female)

install.packages("MatchIt")

library(stargazer)
vars_used <- c("amountdef", "liqwealth", "taxincome",
               "healthexp", "edu", "age", "female")
stargazer(loan_data[, vars_used],
          type = "text",
          title = "Summary Statistics of Main Variables",
          digits = 2)
```

Table 1: Summary Statistics of Main Variables

Statistic	N	Mean	St. Dev.	Min	Max
amountdef	2,000	1.13	3.60	0.00	30.99
liquealth	2,000	14.31	5.92	3.24	37.43
taxincome	2,000	87.82	25.61	27.44	193.65
healthexp	2,000	12.03	5.92	1.83	49.02
edu	2,000	9.10	2.74	5	20
age	2,000	60.28	8.56	34	98
female	2,000	0.50	0.50	0	1

Annex 2: Histogram of loan default

```
hist(loan_data$amountdef,
     breaks = 30,
     col = "lightblue",
     main = "Distribution of Loan Defaults",
     xlab = "Amount of Default (in thousands)",
     ylab = "Number of Borrowers")
```



Annex 3: Summary statistics by default group

```
library(stargazer)
vars_group <- c("liqwealth", "taxincome", "healthexp", "edu", "age")

group_means <- aggregate(
  loan_data[, vars_group],
  by = list(DefaultGroup = loan_data$high_default),
  FUN = mean,
  na.rm = TRUE)

group_means

stargazer(group_means,
          type = "text",
          summary = FALSE,
          title = "Annex 3 - Mean Characteristics by Default Group",
          rownames = FALSE,
          digits = 2)
```

Table 2: Mean Characteristics by Default Group

Default Group	liqwealth	taxincome	healthexp	edu	age
0	14.61	89.95	12.33	9.25	60.56
1	11.71	69.32	9.46	7.78	57.86

Annex 4: Probit model

```
psm_formula <- high_liqwealth ~ taxincome + edu + age + female + health_burden
```

```
psm_model <- glm(psm_formula,
                 data = loan_data,
                 family = binomial(link = "probit"))
summary(psm_model)

library(stargazer)
stargazer(psm_model,
          type = "text",
          title = "Annex 4 - Probit Model for Propensity Scores",
          dep.var.labels = "High Liquidity (1 = above mean)",
          covariate.labels = c("Taxable Income",
                               "Education (years)",
                               "Age",
                               "Female (1 = female)",
                               "Health Burden"),
          digits = 3,
          no.space = TRUE,
          intercept.bottom = FALSE)
```

Table 3: Probit Model for Propensity Scores

Dependent variable: High Liquidity (1 = above mean)	
Variable	**Coefficient**
Taxable Income	-3.954^{***}
	(0.251)
Education (years)	0.029^{***}
	(0.002)
Age	-0.011
	(0.014)
Female (1 = female)	0.019^{***}
	(0.004)
Health Burden	0.409^{***}
	(0.065)
health_ burden	0.031
	(0.720)
Observations	2,000
Log Likelihood	$-1,074.501$
Akaike Inf. Crit.	$2,161.001$
Note: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$	

Annex 5: Average marginal effects

```
library(margins)
ame_psm <- margins(psm_model)
ame_df <- as.data.frame(summary(ame_psm))
```

```
stargazer(ame_df,
          type = "text",
          summary = FALSE,
          title = "Annex 5 - Average Marginal Effects from Probit Model",
          rownames = FALSE,
          digits = 4)
```

Table 4: Average Marginal Effects from Probit Model

Factor	AME	SE	z	p	lower	upper
age	0.0059	0.0013	4.4706	0.000 01	0.0033	0.0085
edu	−0.0033	0.0043	−0.7824	0.4340	−0.0117	0.0050
female	0.1244	0.0193	6.4588	0	0.0867	0.1622
health_burden	0.0095	0.2189	0.0434	0.9654	−0.4196	0.4386
taxincome	0.0088	0.0004	19.8635	0	0.0079	0.0096

Annex 6: Propensity score distribution

```
library(MatchIt)
m.out <- matchit(psm_formula,
                 data = loan_data,
                 method = "nearest",
                 distance = "probit")

plot(m.out, type = "hist")
title(main = "Propensity Score Distribution by Liquidity Status",
      xlab = "Estimated Propensity Score",
      ylab = "Number of Borrowers")
```

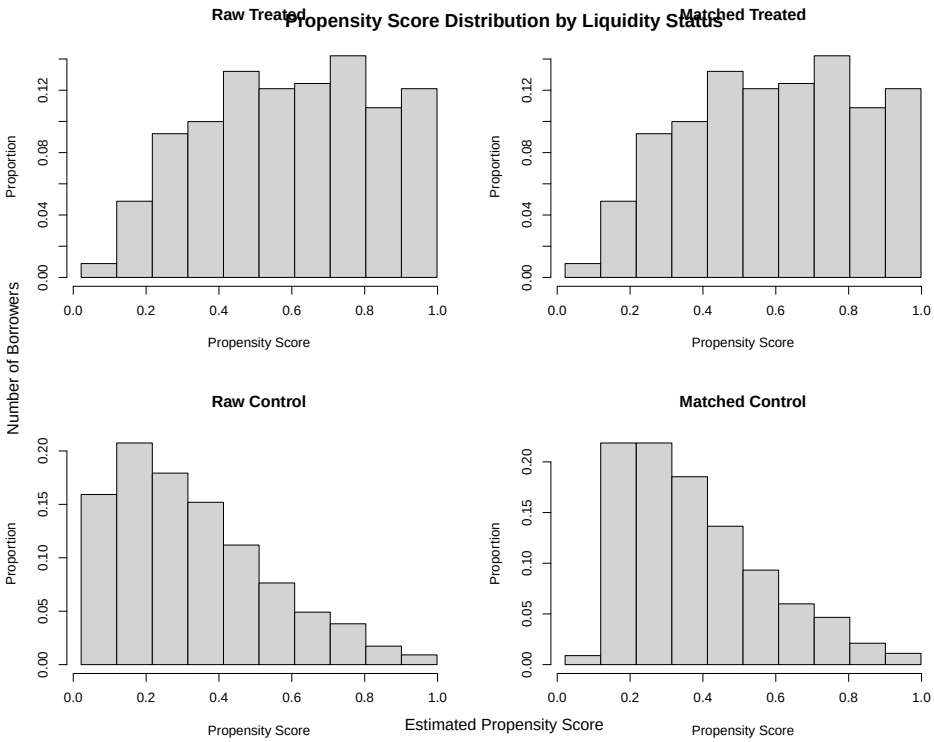


Figure 1: Distribution du propensity score

lov

Annex 7: Covariate balance plot

```
love.plot(m.out,
          stats = "mean.diffs",
          threshold = 0.1,
          abs = TRUE,
          var.order = "unadjusted",
          title = "Standardized Mean Differences Before and After Matching")
```

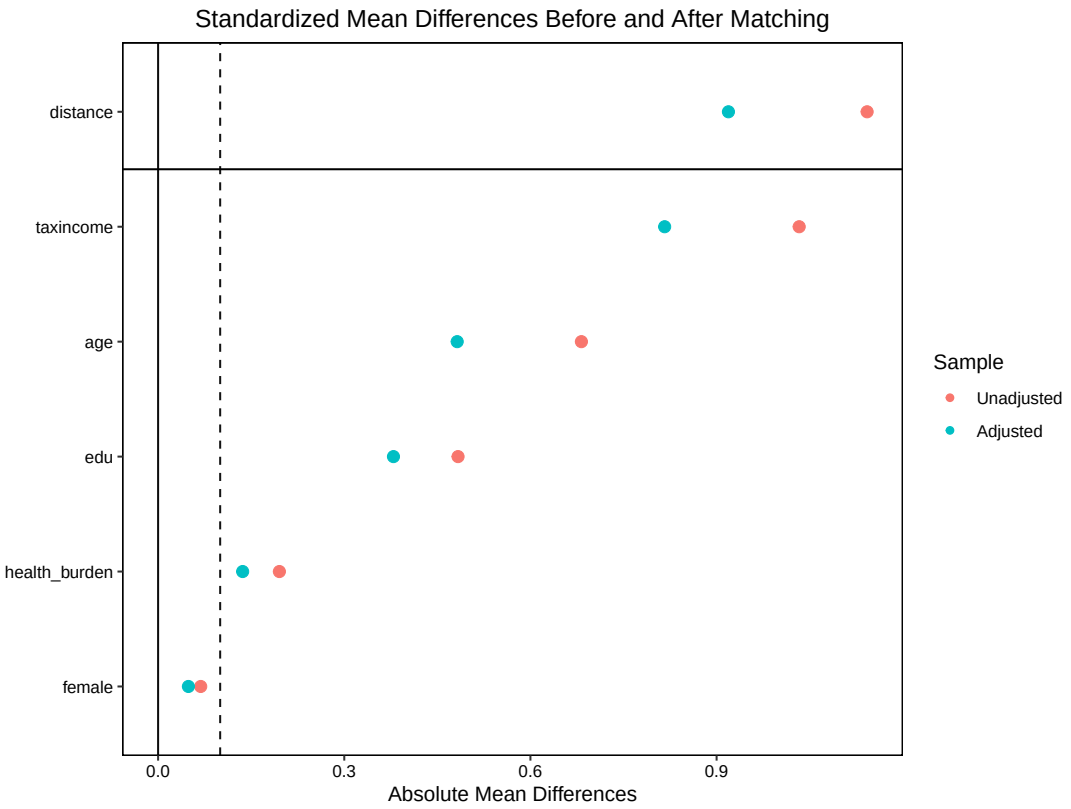


Figure 2: Love plot : équilibre des covariables avant et après appariement

Annex 8: Covariate balance table

```
balance_table <- bal.tab(m.out, un = TRUE)
balance_table

bal <- balance_table$Balance

bal_df <- data.frame(
  Variable = rownames(bal),
  Type = bal$Type,
  Diff_Unmatched = round(bal$Diff.Un, 3),
  Diff_Matched = round(bal$Diff.Adj, 3)
)

bal_df

library(stargazer)
stargazer(bal_df,
          type = "text",
          summary = FALSE,
          title = "Annex 8 - Covariate Balance Before and After Matching",
          rownames = FALSE,
          digits = 3)
```

Table 5: Covariate Balance Before and After Matching

Variable	Type	Diff_Unmatched	Diff_Matched
distance	Distance	1.143	0.919
taxincome	Contin.	1.033	0.816
edu	Contin.	0.483	0.379
age	Contin.	0.682	0.482
female	Binary	0.069	0.049
health_burden	Contin.	0.195	0.136

Annex 9: Naïve OLS vs PSM OLS

```
library(MatchIt)
library(stargazer)

ols_naive <- lm(amountdef ~ high_liqwealth, data = loan_data)

m.data <- match.data(m.out)

ols_psm <- lm(amountdef ~ high_liqwealth,
              data = m.data,
              weights = weights)

stargazer(ols_naive, ols_psm,
          type = "text",
          title = "Annex 9 - Effect of High Liquidity on Default: Naïve OLS vs PSM",
          column.labels = c("Naïve OLS", "PSM (Matched OLS)"),
          dep.var.labels = "Amount of Default",
          covariate.labels = "High Liquidity (1 = above mean)",
          digits = 3)
```

Table 6: Annex 9 – Effect of High Liquidity on Default: Naïve OLS vs PSM

	Dependent variable: Amount of Default	
	Naïve OLS	PSM (Matched OLS)
	(1)	(2)
High Liquidity (1 = above mean)	-0.543***	-0.256
	(0.161)	(0.164)
Constant	1.372***	1.084***
	(0.108)	(0.116)
Observations	2,000	1,802
R^2	0.006	0.001
Adjusted R^2	0.005	0.001
Residual Std. Error	3.590 (df = 1998)	3.490 (df = 1800)
F Statistic	11.347*** (df = 1; 1998)	2.415 (df = 1; 1800)

Note: *p<0.1; **p<0.05; ***p<0.01